1-Dimensional 3-Level

Maxwell-Bloch equations

Scales:

$$t^{\mathrm{SI}} = \tau T_0$$

$$z^{\mathrm{SI}} = \xi L$$

$$E_{32}^{\mathrm{SI}} = \tilde{E}_{32} E_{\mathrm{scale}}$$

$$E_{31}^{\mathrm{SI}} = \frac{\tilde{E}_{32}}{d} E_{\mathrm{scale}}$$

$$I_{32}^{\mathrm{SI}} = \tilde{I}_{32} I_{\mathrm{scale}}$$

$$I_{31}^{\mathrm{SI}} = \frac{\tilde{I}_{31}}{d^2} I_{\mathrm{scale}}$$

$$T_0 = \frac{2\epsilon_0 \hbar}{n_0 L |d_{32}|^2 k_{32}}$$

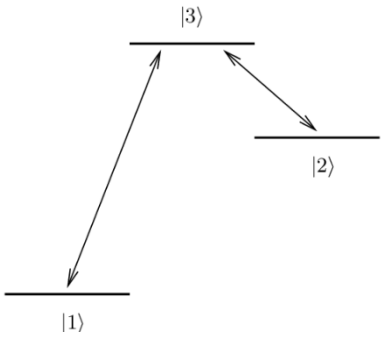
$$E_{\mathrm{scale}} = \frac{i\hbar}{T_0 d_{32}}$$

$$I_{\mathrm{scale}} = \frac{1}{2}\epsilon_0 |E_{\mathrm{scale}}|^2$$

$$d = \frac{d_{31}}{d_{32}}$$

$$\omega = \frac{\omega_{31}}{\omega_{32}}$$

$$\eta = d^2 \omega$$



$$n_3^0 = a n_1^0$$

$$n_2^0 = b n_3^0$$

$$\Rightarrow \begin{cases} n_1^0 = \dfrac{1}{1+a(1+b)} \\ n_2^0 = \dfrac{ab}{1+a(1+b)} \\ n_3^0 = \dfrac{a}{1+a(1+b)} \end{cases}$$

$$0 \leqslant a, b \leqslant 1$$

Bloch Eqs.

$$\dot{n}_{31} = -\operatorname{Re}\left(\tilde{E}_{31}^* R_{31} + \frac{1}{2}\tilde{E}_{32}^* R_{32}\right) - \frac{n_{31} - n_{31}^0}{T_1^{31}} + \Lambda_n^{31}$$

$$\dot{n}_{32} = -\operatorname{Re}\left(\frac{1}{2}\tilde{E}_{31}^* R_{31} + \tilde{E}_{32}^* R_{32}\right) - \frac{n_{32} - n_{32}^0}{T_1^{32}} + \Lambda_n^{32}$$

$$\dot{R}_{21} = \frac{1}{2}\left(\tilde{E}_{31} R_{32}^* + \tilde{E}_{32}^* R_{31}\right) - \frac{R_{21} - R_{21}^0}{T_2^{21}}$$

$$\dot{R}_{31} = \tilde{E}_{31} n_{31} - \frac{1}{2}\tilde{E}_{32} R_{21} - \frac{R_{31} - R_{31}^0}{T_2^{31}}$$

$$\dot{R}_{32} = \tilde{E}_{32} n_{32} - \frac{1}{2}\tilde{E}_{31} R_{21}^* - \frac{R_{32} - R_{32}^0}{T_2^{32}}$$

Initial conditions ($\tau = 0$)

$$n_1^0 = \frac{1}{1+a(1+b)}$$

$$n_{31}^0 = (a-1) n_1^0$$

$$n_{32}^0 = a(1-b) n_1^0$$

$$R_{21}^0 \leqslant 2\sqrt{ab}\, n_1^0$$

$$R_{31}^0 \leqslant 2\sqrt{a}\, n_1^0$$

$$R_{32}^0 \leqslant 2a\sqrt{b}\, n_1^0$$

Maxwell Eqs.

$$\frac{\partial \tilde{E}_{31}}{\partial \xi} = \eta R_{31}$$

$$\frac{\partial \tilde{E}_{32}}{\partial \xi} = R_{32}$$

Boundary conditions ($\xi = 0$)

$$\tilde{E}_{31}(\xi = 0, \tau) = E_{31}^B(\tau) d$$

$$\tilde{E}_{32}(\xi = 0, \tau) = E_{32}^B(\tau)$$

$$\Lambda_n(\tau) = \Lambda_1\left(1 - \tanh^2\left(\frac{\tau - \tau_0}{T_t}\right)\right) \quad , \quad E^B(\tau) = E_0\left(1 - \tanh^2\left(\frac{\tau - \tau_0}{T_t}\right)\right)$$